

ZAYSAN, KazakhstanWMO: **366650**Lat: **47.467N**Lon: **84.917E**Elev: **603**StdP: **94.29**Time Zone: **6.00 (E06)**Period: **86-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.4	-24.9	-30.2	0.2	-26.9	-27.8	0.3	-24.3	10.0	-4.5	7.9	-4.5	1.2	270	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.1	32.0	17.2	30.4	16.9	29.0	16.5	18.7	28.6	18.0	27.7	17.3	26.7	3.4	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.1	11.5	22.2	14.1	10.8	22.6	13.2	10.2	22.3	55.6	28.6	53.2	27.9	51.0	26.5	21.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
10.1	8.0	6.5	-30.0	35.9	3.9	2.5	-32.8	37.7	-35.1	39.1	-37.3	40.5	-40.1	42.3	
			-29.9	20.0	3.7	0.6	-32.6	20.4	-34.8	20.8	-36.9	21.2	-39.6	21.6	
			DB												
			WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.2	-14.6	-14.5	-5.6	7.5	15.5	21.0	23.7	21.7	15.3	7.0	-4.7	-10.9
	DBStd	14.91	5.31	5.77	6.76	5.77	4.76	3.98	3.08	3.60	4.59	5.54	7.45	5.90
	HDD10.0	3282	763	687	485	113	12	0	0	0	9	124	441	647
	HDD18.3	5209	1022	920	742	324	112	19	3	9	111	351	691	906
	CDD10.0	1539	0	0	2	38	181	329	424	363	169	32	1	0
	CDD18.3	426	0	0	0	0	23	98	169	114	21	1	0	0
	CDH23.3	3298	0	0	0	1	157	746	1350	903	136	4	0	0
	CDH26.7	991	0	0	0	0	29	218	430	289	23	1	0	0
Wind	WSAvg	2.5	1.3	1.4	1.5	3.1	3.7	3.5	3.3	3.1	2.9	2.4	1.9	1.6
Precipitation	PrecAvg	301	11	9	14	28	40	31	43	24	25	31	25	17
	PrecMax	483	23	22	37	79	118	76	97	67	82	88	63	45
	PrecMin	150	0	0	0	0	0	0	0	0	0	0	0	4
	PrecStd	75	7	6	10	18	29	18	28	17	16	20	15	10
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-1.0	-0.4	16.1	22.6	29.9	33.0	34.6	34.2	29.4	23.2	13.5	2.1
		MCWB	-2.6	-2.1	9.2	11.8	16.7	17.5	18.4	17.8	15.1	12.9	8.2	0.2
	2%	DB	-4.5	-3.0	9.8	21.0	27.1	30.7	32.1	31.9	26.7	19.9	8.6	0.2
		MCWB	-5.5	-4.6	5.5	11.3	15.2	16.7	17.6	16.7	14.5	10.7	5.6	-1.1
	5%	DB	-6.3	-4.6	6.2	18.6	24.9	29.3	30.5	29.7	24.8	17.4	5.4	-1.2
		MCWB	-7.1	-5.9	2.9	10.1	14.3	16.3	17.3	16.3	13.7	9.5	2.5	-2.3
	10%	DB	-7.7	-6.3	2.9	16.3	23.1	27.6	29.1	28.0	22.6	14.9	3.2	-2.9
		MCWB	-8.4	-7.3	0.6	9.1	13.6	15.9	17.1	15.9	12.8	8.2	1.1	-3.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-2.3	-1.7	9.2	13.2	18.0	19.0	19.7	19.1	16.0	13.4	8.6	0.7
		MCDB	-1.1	-0.5	15.1	20.6	28.1	30.3	28.7	30.3	26.8	22.1	12.5	1.7
	2%	WB	-5.4	-4.3	5.9	11.7	16.5	17.7	19.0	17.8	15.0	11.7	5.9	-1.1
		MCDB	-4.7	-3.0	9.6	19.6	24.6	27.8	28.6	28.1	25.0	18.6	8.9	0.0
	5%	WB	-7.1	-5.9	3.1	10.6	15.2	17.0	18.4	17.0	14.2	9.9	2.8	-2.2
		MCDB	-6.5	-4.7	6.1	18.1	23.2	26.5	27.9	27.3	23.0	16.6	5.1	-1.2
	10%	WB	-8.4	-7.3	0.5	9.2	14.2	16.4	17.8	16.4	13.4	8.6	1.2	-3.9
		MCDB	-7.7	-6.2	2.7	16.1	21.8	26.0	27.1	26.4	21.8	14.5	3.1	-3.1
Mean Daily Temperature Range	5% DB	MDBR	6.2	7.0	6.8	9.1	9.1	8.7	8.1	8.4	9.4	8.5	6.5	5.9
		MCDBR	7.8	7.9	9.3	12.1	11.9	10.1	9.7	10.4	12.1	11.1	9.4	6.8
		MCWBR	6.8	6.8	6.3	6.9	6.5	4.4	4.0	4.4	5.8	6.2	6.7	5.7
	5% WB	MCDBR	7.7	7.9	9.2	11.2	10.0	9.0	8.4	9.4	10.6	10.5	9.0	6.7
		MCWBR	6.8	6.8	6.4	6.9	6.0	4.3	4.2	4.5	5.3	6.2	6.6	5.8
Clear-Sky Solar Irradiance	taub	0.293	0.327	0.379	0.427	0.411	0.417	0.420	0.392	0.358	0.351	0.325	0.295	
	taud	2.138	2.097	2.064	2.057	2.167	2.230	2.245	2.300	2.342	2.255	2.189	2.134	
	Ebn at Noon	761	813	825	822	853	849	839	849	847	781	713	702	
	Edh at Noon	87	113	138	154	143	136	132	120	104	96	83	77	
All-Sky Solar Radiation	RadAvg	1.79	2.86	4.12	5.17	6.48	6.68	6.39	5.91	4.63	2.83	1.61	1.33	
	RadStd	0.09	0.12	0.28	0.35	0.36	0.38	0.49	0.30	0.32	0.27	0.14	0.10	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page